

RYAN RIVELL

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SUMMARY

Mechanical engineer graduating from UT Arlington in December 2026 with over two years of experience at Lockheed Martin spanning mechanical design and systems engineering. Experienced in CAD (Creo, SolidWorks, Onshape), GD&T, tolerance analysis, FEA, and mechatronics, and taking hardware from concept through release. Passionate about designing, analyzing, and building mechanical and electromechanical systems that solve real-world problems.

EDUCATION

Bachelor of Science, Mechanical Engineering

The University of Texas at Arlington

GPA: 3.969

Expected Dec 2026

Arlington, TX

Honors/Awards: Presidential Honors Merit Scholarship, 2024–2025 CWEP Student Employee of the Year (Lockheed Martin/UTA)

PROFESSIONAL EXPERIENCE

Mechanical Design Engineering - College Work Experience Program (CWEP)

Lockheed Martin

Apr 2025 – Present

Grand Prairie, TX

- Own mechanical components and assemblies through design, tolerance analysis, engineering documentation, multidisciplinary review, and release.
- Design structural components, electronics mounting hardware, fixtures, and test hardware in Creo, considering packaging, assembly, manufacturability, structural support, and harness clearance.
- Create and release 2D drawings and Model Based Definition (MBD) models, applying GD&T to define functional interfaces and manufacturing requirements.
- Develop Root Sum Square (RSS) and worst-case tolerance stacks to evaluate design changes against existing analyses and verify assembly fit and dimensional compatibility.
- Perform virtual fit checks and interference analyses; collaborate directly with manufacturing and machine-shop personnel to resolve integration and producibility issues.
- Design and fabricate fixtures, prototypes, mass simulators, and other test hardware using conventional and additive manufacturing methods.
- Developed VBA automation for Creo mass-property updates and Excel data synchronization, reducing repetitive data entry and manual error.

Systems Engineering - CWEP

Lockheed Martin

Apr 2024 – Apr 2025

Grand Prairie, TX

- Contributed to the modernization of systems architecture by developing and modeling a library of components in Cameo Systems Modeler.
- Read and interpreted component specifications to write and refine requirements, developing standardized component requirement specification templates.
- Collaborated with subject matter experts to gather and validate component data, improving model accuracy and traceability across the system architecture.
- Led the initial modeling team in component model publication and contributed to exceeding quarterly milestones, maintaining delivery half a quarter ahead of schedule.

RELEVANT PROJECTS

FSAE Four-Wheel Steering System — Senior Design — *SolidWorks, Onshape, KiCAD, C (programming)*

- Derived actuator torque and speed requirements through physical testing and analytical calculations; performed steering kinematic analysis to develop the control relationship for rear-wheel steering.
- Designed a worm-driven ball-screw linear actuator and gearbox in SolidWorks and Onshape; selected motor, gearing, bearings, seals, and hardware based on load, speed, packaging, and manufacturability.
- Designed system electronics and a custom PCB in KiCAD and developed embedded control software in C using steering-input and actuator-position feedback.
- Performed tolerance and fit analyses for bearings, shafts, and alignment; produced mechanical drawings and a resin 3D-printed functional actuator prototype for controls development and integration.

Hoist Gearbox Design — *Machine Design, MATLAB, SolidWorks*

- Designed a spur-gear reduction gearbox to meet target torque, speed, and service-life requirements using AGMA-based methods; developed MATLAB calculations for gear geometry and sizing and produced the assembly and engineering drawings.

Automotive Scissor Jack — FEA — *ANSYS Mechanical, ANSYS Workbench, SolidWorks*

- Modeled and analyzed a scissor jack assembly under varying loads, applying mesh controls and contact definitions to evaluate stress distribution and safety factors on critical load-bearing components.

SKILLS

Creo, SolidWorks, Onshape, Windchill, GD&T, Tolerance Analysis, FEA (ANSYS), MATLAB/Simulink, C, Mechatronics, Additive Manufacturing

PERSONAL INTERESTS

Automotive, Racing Enthusiast, Space, Robotics, Fitness & Endurance Sports, Guitar